



An introduction to Extended Events

Are you ready to dive into the awesome world of MSSQL Extended Events! In this presentation, we'll explore this great feature in Microsoft SQL Server that helps you monitor and troubleshoot database performance.

We'll start by understanding what Extended Events are and when they come to the rescue. You'll learn how to set them up step-by-step, customizing them to capture the data you need and show you how to look into the results.

We'll also compare Extended Events with the traditional SQL Server Profiler, so you'll know which one to pick for different situations. Plus, we'll share some nifty scripts to start with them right away.



Peter Kruis

SQL Server Specialist

 <https://www.linkedin.com/in/peter-kruis>

 peterkruis@hotmail.com

 <https://github.com/peterkruis>





SeaQL.nl

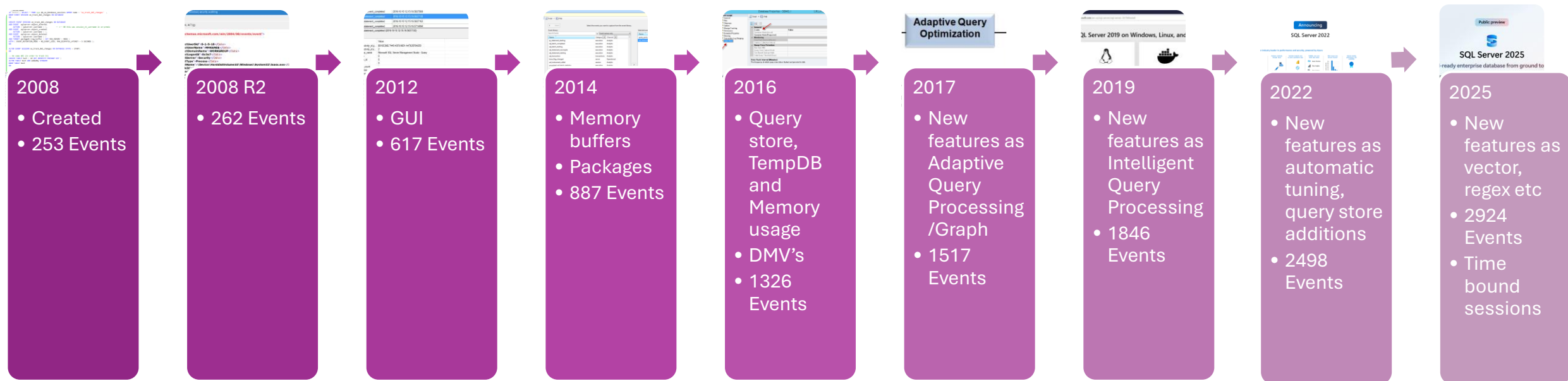
Data, DevOps & SQL @ Sea!

To-be-announced 2027, Ouddorp,
The Netherlands

Why does XE matter?

- Sometimes you need to see what SQL Server is really doing.
- Profiler made tracing accessible
- Extended Events takes that same goal further:
 - Lower overhead
 - More control
 - More events
 - Support for newer SQL Server features

Development



SQL Server 2022

Event	Description
query_antipattern	Fires when an antipattern query has been detected
query_abort	Fires when a query has been aborted
plan_not_cached	Fires when a plan cannot be cached
metadata_changed	Fires when any metadata change operation has been executed
spills_to_tempdb	Fires when memory has been paged to tempdb
parameter_sensitive_plan_optimization_skipped_reason	Fires when a plan cannot use PSPO
And more	About 500 ones 😊

SQL Server 2025

Addition	Description
Time bound sessions	You have the option to set a maximum duration with the MAX_DURATION option.
New functionality (e.g.)	AI Functions
	Vector search / vector index
	Regex
	Newer IQP scenarios

When to choose Profiler



Item	SQL Server Trace	SQL Server Extended Events
GUI Pre-2012	Classic GUI	No GUI, Query should be done through XML
GUI	Classic GUI	GUI with options as Grouping, Filtering, Differences between grid and details
Overhead	Synchronous capture Much more overhead	Asynchronous capture Less overhead
Server/Client	Only for you	Everybody can see and use your trace
Events	No new ones since 2012, so stuck at 180 events	Added each version of SQL (SQL Server 2025 has almost 3000 events!) Azure
Causality Tracking	No ☹️	Yes! 😊
Configure by SQL	More complex syntax	Better granular control
Built-in Replay	Capture and replay	You need to use other tools
Combine with perfmon	Available	Not available

Enough sheets for now..

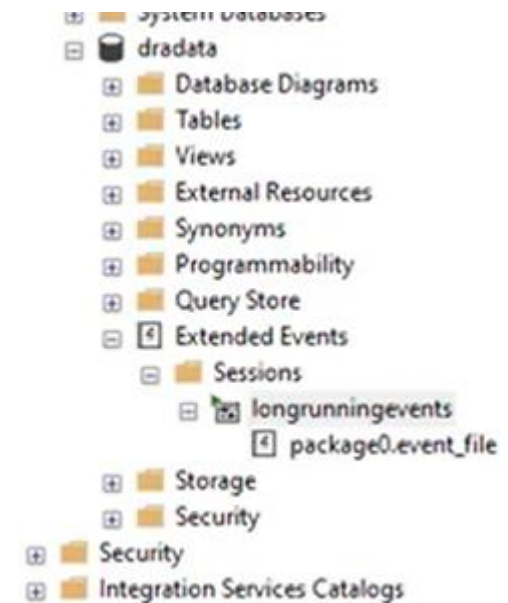
Let's see if I pleased the demo gods enough..

Checklist if we've seen all

- **ClickOPS**
 - Quick start of default XE Profiler
 - New Session
 - Templates
 - Schedule options
 - Causality tracking
 - Events (search, configure, actions, predicate, event fields)
 - Targets
 - Advanced
- **Scripting**
 - New session
 - Results

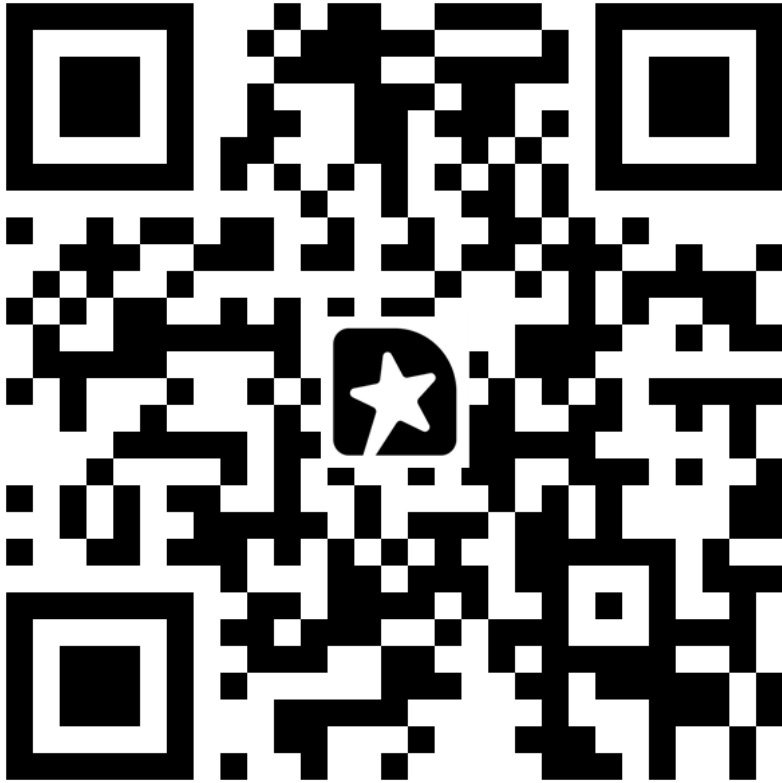
Azure SQL DB: What's different?

- Scope
 - Just database scope
 - CREATE EVENT SESSION ... ON DATABASE
- Target
 - Different types
 - Azure storage blobs only
 - Ring buffer capped
- Location within SSMS



Recap

- Extended Events keep on evolving, so it's the way to go..
- We have seen
 - How can we set it up
 - How can we look into the results
 - Slightly differs on Azure SQL DB



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Feedback please

 <https://www.linkedin.com/in/peter-kruis>

 peterkruis@hotmail.com

 <https://github.com/peterkruis>